

SYE6053: BUSINESS PROCESS IMPROVEMENT AND INNOVATION

Effective Term

Semester B 2025/26

Part I Course Overview

Course Title

Business Process Improvement and Innovation

Subject Code

SYE - Systems Engineering

Course Number

6053

Academic Unit

Systems Engineering (SYE)

College/School

College of Engineering (EG)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

SEEM6053 Business Process Improvement and Innovation (offered until 2021/22) / ADSE6053 Business Process Improvement and Innovation (offered until 2023/24)

Exclusive Courses

Nil

Part II Course Details

Abstract

The course aims to instill students, the concept and value of process improvement by applying quantitative as well as qualitative techniques. Students will learn to appreciate that in today's organizations, process improvement requires analysis with tools; and the capability to improve outcomes with systematic innovative thinking to meet global competitions. This course uses analytical techniques in combination with qualitative tools to let students experience the how's and what's of business process improvement and its role in an organization's long term development.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Explain what is a process in the context of contemporary Organization	10			
2	Discuss the key elements of qualitative and quantitative process analysis and improvement methodology	30	x		
3	Distinguish the concept and value of innovation techniques	20	x	x	
4	Discuss the key quantitative and qualitative methodologies of systematic innovation techniques for process improvement	30	x	x	
5	Discuss the integration and application of innovation techniques with other management tools	10	x	x	

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Lecture, class discussion, Q&A	1, 2, 3, 4, 5	39 hours/sem

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("-" for nil entry)	Allow Use of GenAI?
1	Group Work	3, 4, 5	15	-	No
2	Individual coursework	1, 2, 3, 4, 5	35	-	No

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

2

Minimum Continuous Assessment Passing Requirement (%)

30

Minimum Examination Passing Requirement (%)

30

Assessment Rubrics (AR)

Assessment Task

Group Work (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Quality of submitted group works

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Individual coursework (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Individual submission

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Quality of answers

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Group Work (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Quality of submitted group works

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate/Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Individual coursework (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Individual submission

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate/Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Quality of answers

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(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate/Basic

Failure

(F) Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

- Business process modelling (qualitative and quantitative)
- Business process analysis (qualitative and quantitative)
- Queuing-theoretical analysis of processes
- Process simulation
- Process streamlining
- Value analysis
- Cause and effect chain analysis
- Systematic innovation

Reading List**Compulsory Readings**

Title	
1	Course Notes

Additional Readings

Title	
1	Ravi Anupindi, Sunil Chopra, Sudhakar D. Deshmukh, Jan A. Van Mieghem, & Eitan Zemel, "Managing Business Process Flows, Principles of Operations Management" Pearson Prentice Hall.
2	Darrell Mann Hands on Systematic Innovation IFR Press,2007
3	David Bustare, Peter Kawalek, Mark Norris, editors, "Systems Modeling for Business Process Improvements" Artech House, 2000
4	Manuel Laguna, Johan Marklund , "Business Process Modeling, Simulation and Design", Chapman and Hall/CRC 2018